Nehal Choudhary

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Education

University of California San Diego — **Expected 2027**
B.S. Mathematics-Computer Science, B.S. Cognitive Science-Machine Learning — *GPA: 3.9/4.0*

- Relevant Coursework: Mathematics of Machine Learning (MATH 152), Design & Analysis of Algorithms, Advanced Data Structures, Probability, Linear Algebra, Data Analysis & Inference

Experience

AI & ML Device Engineering Intern | *EchoStar / Boost Mobile* — June 2026 - Present

- **Designed and implemented an AI-powered full-stack Android test automation platform** using Python, FastAPI, React/TypeScript, Appium, and Claude Sonnet through Amazon Bedrock, converting plain-English telecom tests into reusable workflows and reducing manual effort by **80%+**.
- **Engineered an MCP-based LLM planning and validation pipeline** to generate, test, and cache reusable automation flows, enabling reliable execution across Android devices and chipset variations without repeated model calls.
- Scaled concurrent scheduling and test execution across **32+ phones and 110+ scenarios**, adding conflict handling, disconnect recovery, and step-level diagnostics; uncovered a critical **Samsung-to-Samsung RTT interoperability defect** and earned **2nd place** at EchoStar's AI Demo Fair.

Undergraduate Researcher, Multimodal AI | *Cognitive Development Lab @ UCSD* — December 2025 - Present

- Developed a reusable Python evaluation pipeline for Whisper and YOLOv8 across **12 infant-caregiver recordings**, standardizing preprocessing, metric computation, experiment tracking, and result logging.
- Benchmarked Whisper on **2,665 multilingual speech segments** and YOLOv8 across the same dataset, automating failure analysis and achieving **0.912 mean face-detection precision**.

LLM Evaluation Intern | *Netcracker Technology* — July 2025 - September 2025

- Developed an automated **LLM testing and evaluation framework** covering **120+ scenarios and 6 reliability metrics**, replacing manual reviews with repeatable regression tests and structured result reporting.
- Implemented configurable **A/B testing across 20+ prompt variants**, detecting hallucination, scope drift, and schema failures and translating results into engineering recommendations.

Projects

ClassGraph, 1st Place *Python · React · Go*

- Modeled **1,000+ UCSD courses and 100+ prerequisite chains** as directed graphs, using recursive validation and graph traversal to identify scheduling constraints and degree-path conflicts.
- Implemented academic-history PDF uploads, flexible course search, OR-grouping, and duplicate-node handling, improving prerequisite-graph accuracy and usability.

Neural Time-Series Forecasting Hackathon, 2nd Place *PyTorch · GRU · LSTM · Temporal Models*

- Trained and compared **GRU, LSTM, and TCN models** for 10-step neural-signal forecasting across **80–240-channel** multivariate datasets.
- Optimized an LSTM forecasting pipeline with delta-based supervision, teacher forcing, and normalization, reducing MSE from **300K to 50K**.

WildScan, 2nd Place *Python · PyTorch · FastAPI · React*

- Built a mobile-friendly React/FastAPI application that routes image and audio uploads across wildlife classifiers and returns species identifications with LLM-generated threat guidance.
- Trained and evaluated five CNN classifiers using ResNet, EfficientNet, ConvNeXt, and audio CNNs, achieving **71–90% test accuracy** and **100–300 ms image latency**.

Skills and Tools

Languages: Python, C++, Java, SQL, JavaScript/TypeScript, Go
ML & AI: PyTorch, scikit-learn, CNNs, LSTM/GRU/TCN, YOLOv8, Whisper, Amazon Bedrock, Claude Sonnet
Frameworks & Infrastructure: FastAPI, React, REST APIs, Docker, AWS, MongoDB, Git
Automation & Testing: Appium, ADB, Model Context Protocol (MCP), Test Automation, Model Evaluation